Manan Sharma

MCA (AI & ML) — MIT-WPU Pune

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Career Objective

Backend-focused MCA student (AI & ML) with production experience in Python, FastAPI, and Flask — REST APIs, relational schema design, asynchronous pipelines, and automated testing. Experienced in containerised cloud deployments and real operational problem-solving: shipped a compliance SaaS in production and a deployed developer tool. Seeking software engineering and applied AI roles.

Education

MIT World Peace University — **Pune, Maharashtra**
Master of Computer Applications (MCA) — AI & ML Electives | CGPA: 7.60 — *08/2025 – 05/2027*

ICFAI University — **Jaipur, Rajasthan**
Bachelor of Computer Applications — AI & IoT | Percentage: 74.50% — *07/2022 – 05/2025*

Technical Skills

Languages: Python, SQL
Backend & APIs: FastAPI · Flask · REST API Development · Celery · Redis · Pydantic
Databases & Testing: PostgreSQL · SQLAlchemy · Flask-Migrate · Pytest
Cloud & DevOps: Docker · Docker Compose · AWS (EC2, S3, RDS) · Git · GitHub
AI & ML Data: GenAI / RAG Basics · Pandas · NumPy · CatBoost · Streamlit · Statistics

Projects

MineralLaw.in — Mining Compliance Platform (Co-Founder) | *FastAPI, Flask, PostgreSQL, Docker, AWS, Celery, Redis* **2026 –**

- Tracks statutory compliance deadlines for mining lease holders and sends automated WhatsApp reminders ahead of each due date.
- Containerized application stack using Docker Compose; deployed asynchronous Celery workers and managed PostgreSQL database instances on AWS (EC2, RDS).
- Built a rule-based deadline calculator driven by a central configuration of statutory rules, with fingerprint-based idempotency so repeated pipeline runs never create duplicate events.
- Implemented an asynchronous reminder pipeline on Celery Beat with timezone-correct scheduling, and a delivery task guarded by consent checks, a kill switch, and database-level duplicate constraints.
- Designed for data minimisation: phone numbers stored masked in delivery records, with a paginated and filterable admin delivery log for auditing.
- Covered by a 301-test Pytest suite across unit and integration layers, with schema changes version-controlled through Flask-Migrate.

Tokenmeter — LLM Inference Cost Calculator | *JavaScript, HTML, CSS — zero dependencies* — **2026**

- Shipped a single-page tool computing monthly LLM cost across six model tiers with live recomputation, budget-based recommendation, and USD/INR localisation — pricing defaults verified against provider pages; three hand-written files, no frameworks.
- Fixed a spec-level focus-loss race in live table reordering via caret-preserving re-renders and state-guarded input validation; verified in headless Chromium, with full keyboard accessibility and prefers-reduced-motion support.

Food Delivery Time Prediction | *Python, CatBoost, Streamlit*

- Delivery-time regression model trained on 45,593 orders; engineered geo features (haversine and Manhattan-proxy distance, distance buckets) and repaired invalid coordinates using a city-wise median strategy.
- CatBoost reached MAE 5.70 minutes and RMSE 7.26 (R^2 0.40) on a held-out 20% test split — a 7.3% RMSE improvement over a linear-regression baseline and 4.4% over Random Forest.
- Deployed as a Streamlit application performing live ETA prediction with input and coordinate validation.

Certifications

Machine Learning with Python | *IBM / Coursera* — October 2024

Mastering Data Structures & Algorithms using C and C++ | *Udemy — Abdul Bari* — August 2026 – Present

100 Days of Code: The Complete Python Pro Bootcamp | *Udemy — Dr. Angela Yu* — Feb 2026 – Jul 2026

Achievements & Activities

-Identified the MineralLaw.in opportunity independently — existing compliance portals serve regulators and large operators, not small lease holders — then defined the product scope and built it from an empty repository.
-Built and presented working prototypes in two-person teams under time-boxed hackathon conditions, including YOLO Pune 2026 at MIT-WPU.